

Classifying Antarctic Landscapes using Observational Data Spectral Analysis (ODSA)

paper 1 for BedSAT: Antarctica

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Abstract

The subglacial topography of Antarctica is one of the strongest controls on ice-sheet evolution. Nevertheless across most of the continent it has only been observed indirectly along sparse radio-echo sounding (RES) flight lines. This leads to a problem: the topography we want to describe is three-dimensional, but outside a few swath radar-surveyed areas we mostly observe the subglacial landscape in only one dimension, along whatever headings the aircraft happened to fly. We propose that to bound subglacial landscape class it is necessary to have multiple measured statistics that characterise different landscape properties. Here we introduce Observational Data Spectral Analysis (ODSA), an integrated spectral and morphometric framework that combines RES transects with contemporary surface data (REMA, MEaSURES). ODSA builds a vector for landscape classification. Our framework matches the vector to a literature informed catalogue of landscape archetypes based on classifier elements and descriptors. We use ODSA to characterise seven Antarctic regions as proof of concept, quantifying how well the flight lines sample their survey footprint. We compare the results for transects in these regions to their published landscape classes [1] and find them to be distinct but complementary, since ODSA returns a bounded set of admissible archetypes per transect window where the continent-wide product carries one class per cell. No element separates those classes on its own, and the overlaps are too large for better precision to fix, so the bound needs several elements together. Because each vector element is a number rather than a label, a classified landscape can empirically inform synthetic bed topography for the next generation of ice-sheet models.

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Scope

1.1 Aim and Hypothesis

The development of a physically informed spectral analysis framework for quantifying Antarctic subglacial landscape characteristics and classifying bed morphology into archetypes for ice sheet modelling.

We propose that to identify subglacial landscape class it is necessary to have multiple measured statistics. A radar echo sounding (RES) transect bounds the class rather than pinpointing it. Our scheme uses a vector whose elements are either used as classifying axes or descriptors to match RES transects into landscape archetypes defined from the literature. Beds usually fit more than one label and the classifiers typically leave 2, descriptors check whether a surviving label is covering measurably different beds. Our transect analysis scheme recovers bed texture directly along the radar profile, where bed texture 50 km apart decorrelates. By contrast, current continent-wide gridded products such as Bedmap3 is posted at 500 m, but 93% of its ice-thickness cells are interpolated between flight lines [2], so texture at these scales is inferred rather than measured.

1.2 Objectives

- **Derive a landscape classification vector** based on spectral and morphometric elements
- **Landscape archetype mapping:** Taxonomic classification integrates spectral metrics of subglacial terrain into distinct literature-derived landscape archetypes using multiple variables.
- **Application:** The framework establishes parameter bounds for realistic synthetic bed topography generation by incorporating information such as data coverage and migration artefacts, while taking into account possible degeneracy between classes.

218 words

Introduction

Understanding the history and evolution of the Antarctic Ice Sheet (AIS) requires knowledge of Antarctic bed topography, since it is one of the most crucial boundary conditions constraining AIS dynamics. Unfortunately, the direct observation of subglacial landscapes is non-trivial since these are buried under kilometres of ice. Currently, our ability to observe subglacial terrain relies mostly on indirect methods such as RES and seismic sounding, which return 1D profiles of the bedrock; gravity and magnetic surveys, which constrain composition and basin geometry in 2D; and swath radar, which returns 2D bedrock topography. Even though swath radar represents a major improvement over conventional radar sounding, the typical survey geometries (with line spacings ≥ 5 km and widths of only a couple of kilometres) still yield incomplete coverage and are subject to data degradation by interpolation and down-sampling [3]. Contemporary maps of the Antarctic topography fall into the following categories:

Interpolated grids, such as a continent-wide product Bedmap3, which is posted at 500 m, but this is not its resolution, the along-track sampling is metres to kilometres while across-track sampling is hundreds of metres to hundreds of kilometres, so 93% of its ice-thickness cells are interpolated values [2], so texture at these scales is inferred rather than measured, binding to the recovered bedrock texture by the line spacing rather than the grid posting. Alternatively, **inversion models** do not rely on bed measurements but instead infer bed topography indirectly, using ice surface observations and inverse modelling to exploit the fact that basal variability is transmitted by the ice flow to a glacier surface as a function of wavelength [4]. Inverse methods such as IFPA applied across Antarctica [1] are limited in their spatial resolution, since ice flow over bedrock features shorter than the local ice thickness induces negligible perturbations at the ice surface. That floor is a best case and not the delivered resolution. The IFPA product is presented as a mesoscale map covering 2 to 30 km, with its derived texture metrics: hill count and Fourier fractal dimension computed over wavelengths greater than 5 km and reported on $50 \text{ km} \times 50 \text{ km}$ cells [1]. Lastly, there are **hybrid models**, like BedMachine Antarctica v4 [5], are susceptible to both disadvantages described above. Among its methodological strategies to improve on v3, BedMachine Antarctica v4 interpolates the bed by mass conservation in areas of fast ice surface flow (30 m a^{-1}) and by IFPA in areas below that speed threshold. All these examples leave considerable uncertainty in the fine-scale character of the bed.

It is still uncommon for methods to systematically quantify the relationship between bed topography and its morphometric landscape statistics. Studies that measure the statistics report them as continuous fields and terrain types are assigned to specific regions rather than to individual cells, with each parameter described qualitatively. Siegert et al. [6] define three bed roughness groups across named areas of East Antarctica, and Rippin et al. [7] assign three terrain types to the areas of their survey. Numerical thresholds have been published for single parameters, and Jordan et al. [8] divide the Hurst exponent at 0.5 and 0.75 to define low, medium and high roughness categories, but no scheme applies thresholds to a combination of axes per analysed cell. For published classified maps, each cell is often assigned to one class with no margin reported alongside it: Ockenden et al. [1] map the Antarctic bed into five landscape

classes on 50 km cells, and their accompanying data release [9] records each assignment without a confidence interval or an alternative class. In addition ambiguity in landscape class is often only acknowledged as a caveat or analysed under a qualitative lens of multiple interpretations but it is not rigorously quantified, so a published class cannot be read alongside the alternatives that the measurement might still admit. That absence of margins makes such maps vulnerable to survey artefacts, since local roughness measurement bias due to radar migration status or uneven coverage can move a measurement across a boundary without leaving any trace. This often results in averaged regimes that in turn fail to capture the true continuum of the subglacial landscape.

Here we introduce Observational Data Spectral Analysis (ODSA), a physically informed framework for quantifying Antarctic subglacial landscape characteristics and classifying bed morphology into archetypes for ice sheet modelling. The landscape classification provides an empirical foundation for the generation of structurally realistic synthetic bed topographies. ODSA operates on the along-track picks published with Bedmap3 instead of a gridded product, the tracks are visible in Fig.2.1, measuring the survey data directly and avoiding the interpolation we described earlier.

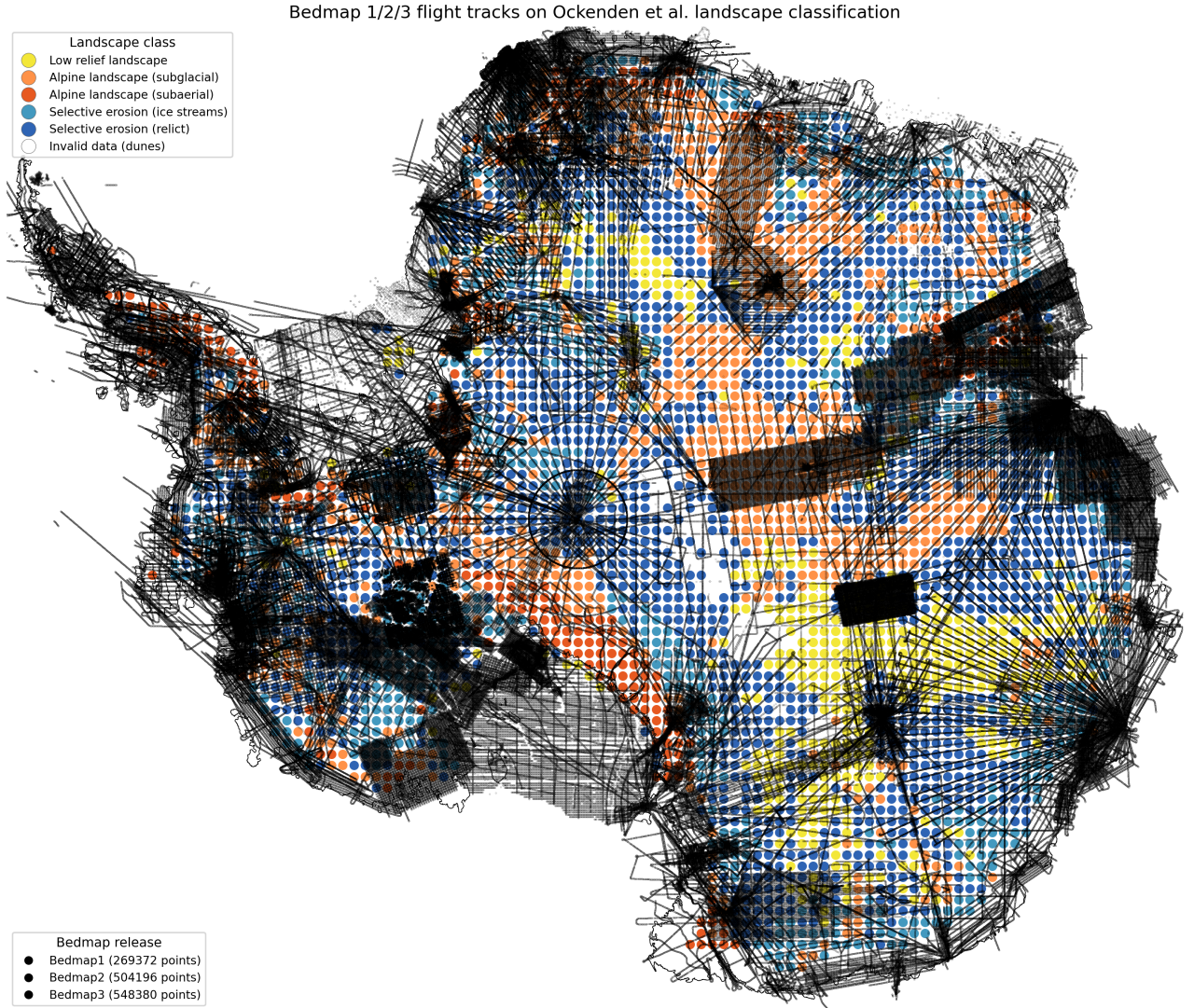


Figure 2.1: Bedmap1, Bedmap2 and Bedmap3 along-track survey points (black) over the continental landscape classification map by Ockenden et al. [1] (coloured cells, cells masked as invalid data (dunes) are drawn white). We reproduce the classification from the published metric grids [9] on 50 km cells. The point counts in the legend show the number of subsampled plotted points rather than published picks. ODSA operates on these along-track data, so the figure shows the sampling available to us across a continental landscape classification.

To prevent physically meaningless parameter averaging within RES transects, ODSA employs bedrock-gradient segmentation and surface-flow confidence weighting to ensure that the extracted metrics represent homogeneous terrain. Since we are dealing with 1D RES transects, ODSA uses 50 km sliding windows as the fundamental unit to spatially analyse the data sets. For a chosen region, the classification scheme is run for each window and is described by an 11-element morphometric vector that combines four classifying axes (classifiers) and seven continuous descriptors. The descriptors are continuous and are reported as numbers, so the morphometry survives the classification rather than being replaced by it. While each classifier carries an uncertainty envelope of two standard deviations at either side of the measurement. ODSA evaluates each envelope against a landscape archetype catalogue with 11 entries and admits every entry the envelope allows. When a window is landscape degenerate the ODSA pipeline names the axis responsible for the overlap and reports the observable required to break the tie. These capabilities impact survey design by telling us which instrument to fly. Additionally bed texture decorrelation between nearby transects constrains the track spacing a survey needs to resolve the landscape it is flown to characterise.

946 words

Methods

1. Data

- All measurements are made on the Bedmap3 point data at the original sample positions [2], rather than on an interpolated grid.
- `processing_flag`: migration status, uniform per CSV (reported only)
- We apply the framework to seven reference regions, which cover low-relief, alpine and selective-erosion landscapes as reported by Ockenden et al. [1]. Five regions use the PS71 boxes from [9]. The Aurora low-relief region is restricted to the low-relief cells of the Ockenden classification, and the Pensacola Pole Basin region uses a box that we defined, with two isolated flight legs equatorward of 88.5°S trimmed.
- The REMA surface elevation model [10] gives ice thickness and the local ice-flow direction, and therefore the orientation of each profile relative to ice flow is also recorded. Ice thickness is the REMA surface elevation minus the Bedmap3 bed elevation. The flow direction comes from the REMA surface gradient with a baseline of 10× the ice thickness following McCormack et al. [11]. Where the elevation difference across the stencil is below $3\sqrt{2} \times 1$ m, the direction is set to undefined.
- MEaSURES: The pipeline uses the 450 m v2 product [12]. At each point it gives the speed and the angle between the MEaSURES velocity direction and the REMA flow direction. These are exported per window as `measures_speed_mean`, `flow_error_mean` and `flow_error_median`. The ERRX/ERRY error fields feed the velocity uncertainty.

2. Processing and Analysis

- ANALYSIS UNIT: 50 km sliding window at 50% overlap (Not every window is 50 km. A segment shorter than 50 km is analysed as a single window the length of the segment)
- Gap cuts (`split_into_segments` module in `segmentation.py`). The track is cut wherever two consecutive points are more than 2000 m apart. Pieces shorter than 50 points or 10 km are dropped.
- Landscape splitting (`split_by_landscape` in `segmentation.py`). After the gap cuts, each piece of track is smoothed with a 50 km running mean, and every point where the smoothed bed gradient exceeds 15 m/km is marked as a transition. Transition zones less than 5 km apart are merged, and the track is cut at the start and end of each zone. Pieces shorter than 50 points or 10 km are dropped. Transition segments go through the same spectral analysis as every other segment, and `is_transition` is written to the window and segment outputs. Transition windows are excluded from classification and from the validation tests. Relief is `elevation_max - elevation_min` of the bed elevations in the window, with no detrending. A change in terrain with a smoothed gradient below 15 m/km is not marked as a transition, so a window can still straddle two terrains, and its relief then includes the elevation offset between them as well as the elevation range within each one. Splitting fallback: if both pieces fail the gates, the parent reverts to one unsplit segment.
- A segment with less than 20% valid ice thickness is skipped.
- Survey grading: `coverage_tags.py`, A/B/C tiers on interpolation distance, azimuth concentration and homogeneous window count, more in Chapter 5

3. Spectral method

- The detrended bed profile in each window is analysed with a Lomb–Scargle periodogram (over 500 geomspace frequencies), which accommodates uneven along-track sample spacing [13]
- Hann taper
- Fit band 250 m to the shorter of segment length and 50 km; the upper edge tracks `WINDOW_SIZE`
- Nyquist floor, $1/(2 \times \max(dx_{\text{median}}, 15 \text{ m}))$, which sets the 30 m short-wavelength limit
- Peak-masked power-law fit. For each segment, the window periodograms are averaged, and a first-pass straight line is fitted to $\log_{10}$ PSD against $\log_{10}$ frequency over the fit band. Every peak at which the averaged PSD is at least twice the fitted value is masked, together with five frequency bins either side. Each window’s own periodogram is then fitted by unweighted least squares over the masked band, which gives the window β , its formal error and $A_{1\text{km}}$; this window β is the one in the vector. The segment β is a refit of the averaged PSD over the same mask, weighted by $1/\text{std}(\log_{10} \text{ PSD})$ across the windows, and unweighted for segments with fewer than three windows. For a single-window segment, the segment β is the window β . The segment β is written to the segment outputs and does not enter the vector, since a segment unit in the vector uses the median of its window values.
- A power law fitted to the spectrum gives the spectral slope β and the spectral amplitude at 1 km wavelength [8, 14]. We also compute the characteristic wavelength [7, 15] and a transect version of the subglacial hill count [1]. We detrend each window, then count a point as a hill if it is the maximum of a 5 km box centred on it and the relief of box exceeds 20 m. Runs of adjacent flagged points count once, and runs that touch a window edge are dropped. Ockenden et al. release the areal count at four gates, 20, 50, 100 and 250 m [9]; we emit the 20 m column, the only one of the four that stays usable in every region here. A one-dimensional transect through a two-dimensional field does not reproduce an areal count, so `hill_count` is our own descriptor built on their recipe rather than a reproduction of their metric.
- Untapered residuals are kept separately for `rms_roughness`, skewness and kurtosis, since the taper would bias a moment
- `xi_band` is the integral of the PSD over the 300 to 1200 m band. `eta_wavelength_m` is calculated over the same band.
- the window formal error exists only where the masked band keeps more than two points

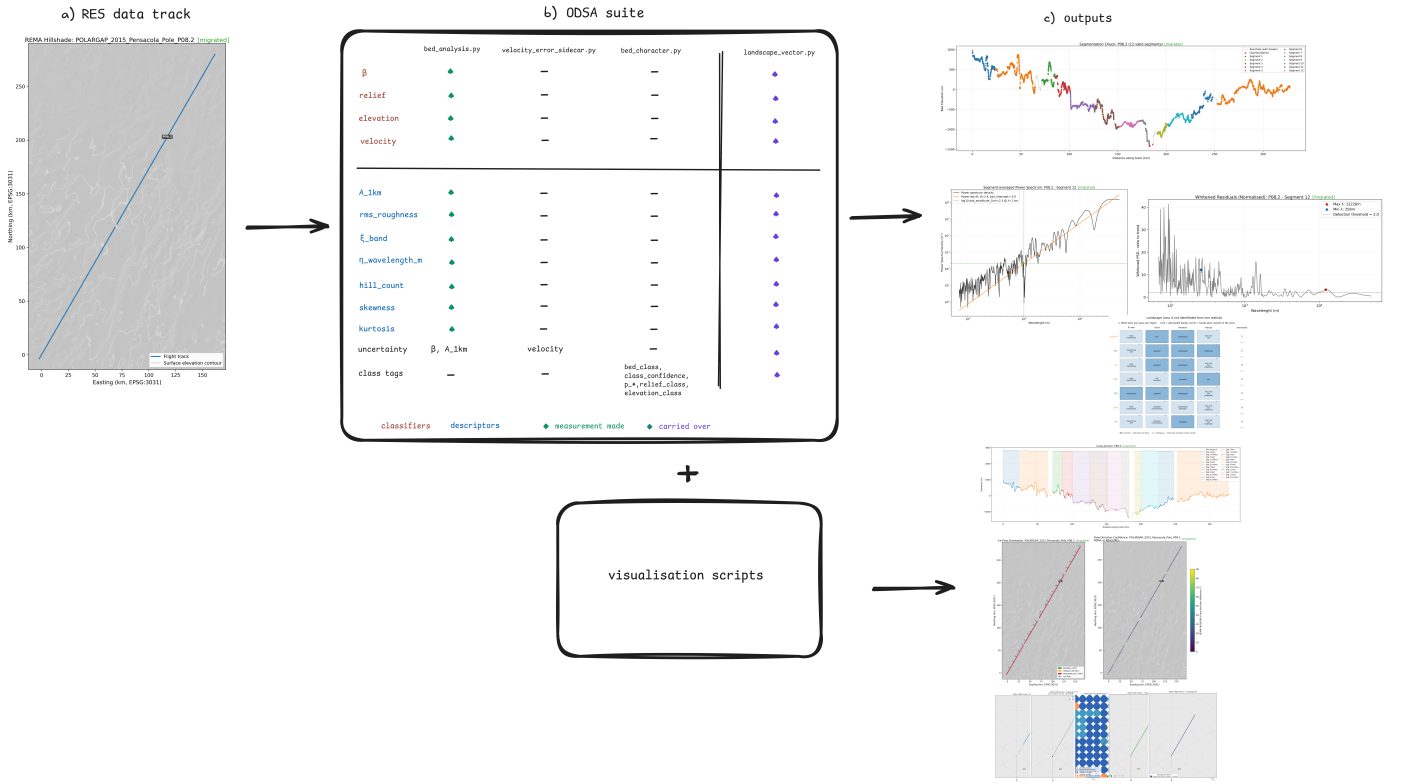


Figure 3.1

4.

3.1 ODSA's Landscape Classification Vector

To inform the synthetic bedrock database that will be used in the training, we need bed morphometric statistics to describe a landscape class. We apply ODSA to seven reference regions as proof of concept (Fig. 3.2), quantifying how well the flight lines sample their survey footprint.

Our proposal is that to bound subglacial landscape class it is necessary to have multiple measured statistics that characterise different landscape properties. ODSA's landscape vector carries 11 elements measured per window. Four classify and seven do not. Each classifier carries an uncertainty envelope of two standard deviations on either side of the measured value, taken from the spectral fit covariance in the case of the roughness slope (β) and from measurement error budgets for relief, velocity and elevation.

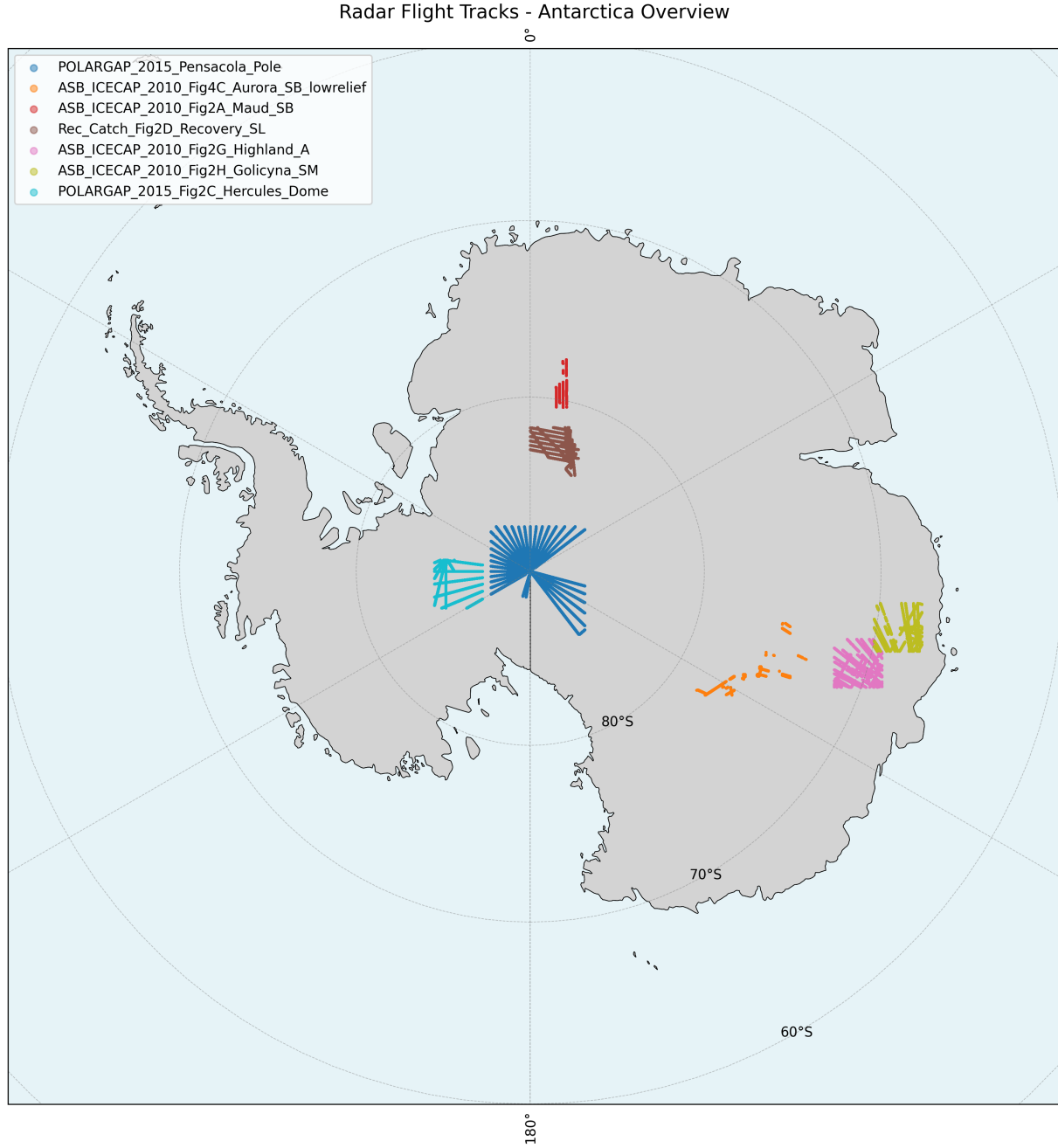


Figure 3.2: The seven RES survey regions we analysed with ODSA. Five regions use the PS71 boxes from [9] directly, while the Pensacola Pole Basin region and the Aurora low-relief region (restricted to low-relief cells of the Ockenden classification) use boxes that we defined.

Table 3.1: Classification axes of the landscape classification vector. Across all four, the framework classifies each axis on value $\pm 2\sigma$, so a window can be admissible in more than one band.

Variable	Description/Thresholds
Classifiers β	Slope of the bed power spectrum over the fitted band (the first of two degrees of freedom obtained by the fit: β and A_1km). A measure of bed roughness shape: low β distributes power to the short wavelengths; high β concentrates it at long wavelengths. The fit band spans 250 m to 50 km. The class thresholds are: chaotic < 1.5 , hard 1.5 to 2.0, transitional 2.0 to 2.5 and soft ≥ 2.5 . The 2.0 and 2.5 breaks follow the Hurst-exponent splits of [8]. The 1.5 break is an operational choice. Beds in the hard class are interpreted as scoured crystalline terrain, like in the areal-scouring landscapes of [16].
relief_m	Elevation range within a window, calculated as $\max(\text{bed}) - \min(\text{bed})$ over the raw 50 km profile, in metres. It gives the amplitude of the largest structure a window contains. Regional slope enters the range directly. Relief class thresholds are set to flat: < 350 m; mountainous: ≥ 800 m; while values in between these are labelled as subdued. These are set based on the median relief for the tested regions.
measures_speed_mean	Mean MEaSUREs InSAR surface speed at the window's track points, sampled over the same points that enter the spectral fit. Bands are very_low (< 5), low (5 to < 10), moderate (10 to < 50) and fast (≥ 50) all in m/yr and set operationally. Antarctic surface speed is bimodal, with a slow mode at 4 to 5 m/yr and a fast mode at 250 m/yr, and measured speed departs from internal deformation above about 15 m/yr [12]. Each window carries a sampled error from the ERRX/ERRY fields of that product [17].
bed_elev_mean	The mean bed elevation within a window, in metres above present-day sea level. Class thresholds are set to submerged: < 0 m; elevated at ≥ 1000 m, and emerged for values in between. Motivated by [6, 7, 18].

5. Uncertainty envelopes - Two standard deviations either side of the measured value, on the four classifying axes only

- β : formal fit error in quadrature with **BETA_SYSTEMATIC_ERROR** = 0.05, a floor rather than an estimate

- Relief 30 m, from $\sqrt{2} \times 20$ m; the ± 7 m surface term is common-mode over 50 km and cancels

- **VELOCITY_ERROR_M_YR** = 5 fires only where a region has no sidecar, which does not happen anywhere in the seven region set

- Elevation 10 m; velocity per window from the ERRX/ERRY sidecar on a 3×3 stencil at ± 15 km, projected onto flow direction above 5 m/yr and isotropic RMS below it

- A failed measurement widens to every band and reads unavailable; it never narrows the admissible set

- The descriptors are continuous because we found no defensible literature threshold.

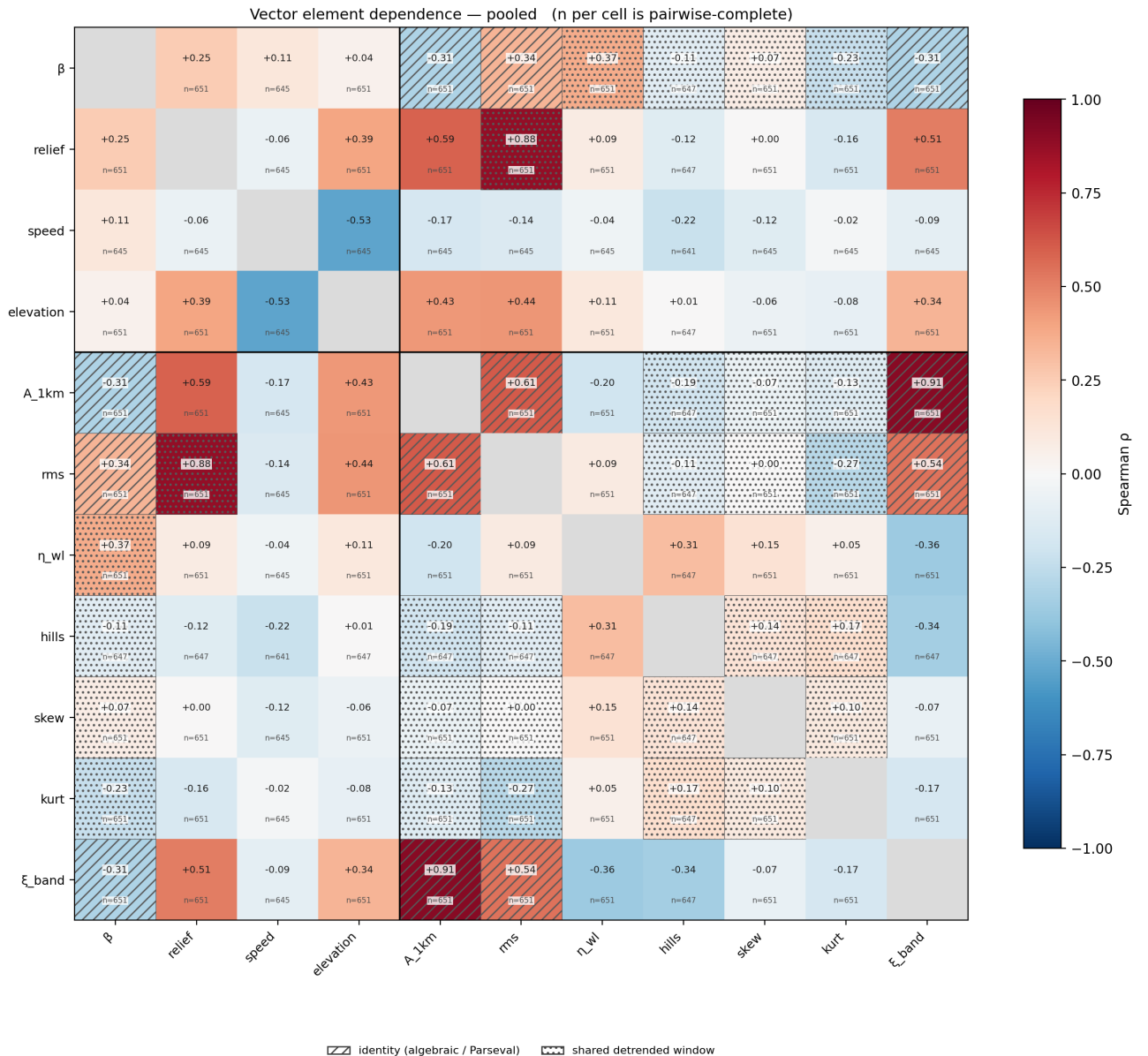
- Dependence structure of vector elements: How much duplicated information do two vector elements (classifiers and descriptors) provide the scheme?

Table 3.2: Descriptors of the landscape classification vector. These are continuous variables with no breaks.

Variable	Description
A_1km	$\log_{10}$ of bed power spectrum at a 1 km reference wavelength evaluated from the fit. It gives the magnitude of the roughness in $\log_{10}(\text{m}^3)$ and fixes the level inside the fitted wavelength band. It carries a propagated uncertainty that includes the slope and intercept covariance. The spectral fit returns two degrees of freedom (β and A_1km) following the conclusion established by [7, 8, 15].
rms_roughness	Root mean square height [19] of the window's detrended elevation residuals, in metres. It gives the amplitude of the topography with the regional ramp removed, which distinguishes it from relief. It depends on profile length, and is retained for comparability with the space-domain roughness literature rather than as a separate degree of freedom.
ξ	Integral of the bed power spectrum over the bedform band from 300 m to 1200 m, in m^2 . ξ is the numerator of η [15]. Supported by [20, 21].
η	Characteristic wavelength for the window, in metres, following [15]. It is the square root of the weighted harmonic mean of the square of the wavelength at each frequency in the band from 300 m to 1200 m with the spectrum as the weight.
hill_count	Number of distinct local maxima in the window's detrended bed profile, similar to the bumpiness metric by [1] but in transect form (1D). A point counts if it is the maximum of a 5 km box centred on it and the box relief exceeds 20 m; adjacent flagged points count once and maxima touching a window edge are dropped. Two hills must be at least 2.5 km apart, so the count cannot exceed 20 per 50 km window.
skewness	Third standardised moment of the window's detrended, untapered bed elevation residuals (bias-corrected). Positive values imply narrow highs on a broad low base, negative values mean narrow lows cut into a broad high surface, following the qualitative hills-against-troughs distinction in [16]. It is the only phase-asymmetry information in the vector.
kurtosis	Excess kurtosis of the window's detrended, untapered bed elevation residuals (bias-corrected). Positive values mean a few extreme highs or lows against a flat background, negative values mean a flat-topped or evenly spread distribution.

Table 3.3: Observables that ODSA cannot supply from RES data. They are required to separate the admissible entries, or to confirm the entry on a row with one admissible entry

External	Description
Reflectivity	Ice-water vs ice-sediment interface (specularity). Required by: BASIN. Separates: water from sediment under a flat, soft, submerged bed. Why ODSA can't measure it: needs raw radargram waveform amplitude.
Composition	Gravity / magnetics (sediment vs smooth bedrock). Required by: BASIN-HIGH. Separates: smooth hard bedrock from soft sediment. Why ODSA can't measure it: needs airborne gravity or magnetics
Flow history	Disrupted internal layers, dating (active vs relict). Required by: TRUNK-RELICT. Separates: TRUNK-RELICT from BASIN. Why ODSA can't measure it: needs disrupted internal layers or dating.
Origin	Erosional vs structural provenance of the trough-ridge contrast. Required by: DISSECTED. Separates: the three possible origins of DISSECTED, and DISSECTED from RIFT. Why ODSA can't measure it: needs independent geological context

**Figure 3.3**

6. Catalogue matching

- The operation is exclusion, not identification.
- 11 entries; an entry is a set of allowed values per axis, and an unconstrained axis can never exclude that entry. An ambiguous axis keeps an entry in.
- Definitions the results depend on: RESOLVED, RESOLVED-WITH-EXTERNAL, DEGENERATE and OUT-OF-CATALOGUE, and the separable flag
- Degeneracy reporting: name the axis responsible and the observable that would break the tie

7. Validation procedures

- Concordance against Ockenden et al. [1]: `ockenden_class.py` reproduces the masks from the Zenodo metric NetCDFs, 50 km cells against 50 km windows, matched-scale join, `MIN_N = 10`, separation criteria $z \geq 2$ and $d \geq 1$, run on all windows and on non-straddling windows only
- Catalogue-blind test at segment level, on the seven descriptors. A single-window segment cannot enter it, since the test needs a within-unit spread on both members.
- Reproducibility: co-located 30 km and 50 km windows within one trajectory; note that β 's figure is a floor because the fit band tracks the window size
- Decorrelation (`window_atom_test.py`). For every pair of non-transition windows, we record whether the two windows share each class (β class, relief class, velocity band and elevation class) and whether they share all four, binned by the distance between window centres. We do this for pairs within one trajectory and for all pairs within a region. The class used here is the class of the measured value, without the uncertainty envelope. Chance agreement is the fraction of randomly shuffled window pairs that share all four classes. The distance at which agreement on all four classes is still about twice chance between 200 and 300 km and reaches chance only beyond 300 km, so `COMPOSITION_DECIMATE_KM = 200` km is a practical floor rather than a demonstrated independence distance, and `n_independent` is an upper bound. We count independent windows by greedily selecting windows with centres at least 200 km apart.
- Within-cell variance ordering: report as ordering only, never as a number or a range

ODSA measures bed statistics from radar data at the original sample positions, therefore the synthetic beds used to train BedSAT will be based on observed Antarctic bed properties rather than on interpolated grids. The continuous elements of the landscape vector provide the morphometric statistics that a synthetic bedrock generator can take as inputs to generate physically informed beds at ice-sheet model resolution. Bed topography is expressed at the ice surface [4, 22]. BedSAT will use ISSM ice flow simulations over these synthetic beds, resulting in a synthetic bedrock and resultant surface expression pair. BedSAT will learn bed-to-surface transfer from these synthetic pairs and invert it, so that satellite observations of the ice surface can be mapped back to bed topography at greater accuracy.

1560 words

Results

ODSA produces four results that advance the characterisation of the Antarctic bed. First, neither β nor relief separates any pair of the published landscape classes [1] on co-located windows, and no element of the vector separates more than half of the class pairs, because the distributions of each element overlap between classes. Landscape classification from radar profiles therefore requires several elements measured together. Second, a radar profile constrains the landscape class without elements measured together. Second, a radar profile constrains the landscape class without identifying it uniquely: of the 11 archetypes in the catalogue, a median of two remain admissible per window over 651 windows. Third, the bed texture elements largely decorrelate within one 50 km window length along a profile, while bed elevation and surface speed remain correlated over hundreds of kilometres, and the bed class of windows on neighbouring flight lines 1 to 25 km apart agrees only slightly more often than expected by chance. Bed texture therefore changes between neighbouring flight lines and cannot be recovered by interpolating between them. Fourth, in every region the detection limit for a difference in β between flow-parallel and flow-perpendicular profiles is two to twenty times larger than the difference measured on deglaciated beds with mapped glacial fabric. Reducing this limit requires one to two orders of magnitude more coverage from flight lines that cross the same ground at several angles.

227 words

Discussion

The scheme is explicit about what the vector cannot do, for example: Material, thermal and hydrological identities do not follow from RES geometry. So where two physically different beds collapse onto one vector the scheme labels it degenerate rather than resolving it by assumption.

5.1 Data Representativeness and Model Utility

5.2 Ockenden landscapes comparison

44 words

Conclusion

words

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